BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: 6197 / 6198

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) Group - I

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6197

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If ${}^{n}C_{8} = {}^{n}C_{12}$, then 'n' equal to:

(A) 4
(B) 8
(C) 20
(D) 12

Q2. The inequality $4^{n} > 3^{n} + 4$ is true for:

(A) $n = 1$
(B) $n \geq 2$
(C) $n = 0$
(D) $n < 2$

Q3. Middle term of $(a + b)^{11}$ is / are:

(A) 6th
(B) 5th & 6th
(C) 6th & 7th
(D) 5th

Q4. $\cot^{2}\theta - \operatorname{cosec}^{2}\theta$ equal to:

(A) 2
(B) 0
(C) 1
(D) -1

Q5. $\tan(\pi - \alpha)$ is equal to:

(A) $\tan \alpha$ (B) $-\tan \alpha$

(C) $\cot \alpha$ (D) $-\cot \alpha$

Q6. The period of $3\cos(x/5)$ is:

(A) π (B) $\pi/10$

(C) 10π (D) $\pi/5$

Q7. If ΔABC is right triangle such that $m\angle\alpha = 90°$, then with usual notations, the true statement is:

(A) $a^2 = b^2 + c^2$ (B) $b^2 = a^2 + c^2$

(C) $c^2 = a^2 + b^2$ (D) $a^2 = b^2 = c^2$

Q8. $a/\sin\alpha = b/\sin\beta = c/\sin\gamma$ is called:

(A) Law of cosine (B) Law of sine

(C) Law of tangents (D) Law of fundamental trigonometry

Q9. $2\tan^{-1}A$ equals:

(A) $\tan^{-1}(2A/(1-A^2))$ (B) $\tan^{-1}(A/(1-A^2))$

(C) $\tan^{-1}(A/(1+A^2))$ (D) $\tan^{-1}(2A/(1+A^2))$

Q10. If $\sin x = \cos x$, then $x =$:

(A) $0°$ (B) $30°$

(C) $45°$ (D) $60°$

Q11. The multiplicative inverse of complex number (0, -1) is equal to:

(A) (1, 0) (B) (0, 1)

(C) (-1, 0) (D) (0, 0)

Q12. The domain of $f = \{(a, 1), (b, 1), (c, 1)\}$ is equal to:

(A) {a, b, c} (B) {a}

(C) {1} (D) {b, c}

Q13. The inverse of a square matrix exists if A is:

(A) Singular (B) Non-singular

(C) Symmetric (D) Rectangular

Q14. The matrix [a b c d] is:

(A) Square (B) Unit

(C) Null (D) Row matrix

Q15. The number of roots of polynomial equation $8x^6 - 19x^3 - 27 = 0$ are:

(A) 2 (B) 4

(C) 6 (D) 8

Q16. If ω is the cube root of unity, then $(1 + \omega - \omega^2)^8$ = :

(A) 256 (B) -256

(C) -256ω (D) 256ω

Q17. Types of rational fractions are:

(A) 3 (B) 2

(C) 4 (D) 1

Q18. Reciprocal terms of an arithmetic progression form:

(A) A.P. (B) H.P.

(C) G.P. (D) Sequence

Q19. If A, G, H have their usual meanings and 'a' and 'b' are positive distinct real numbers and G > 0, then:

(A) A > G > H (B) A < G < H

(C) H > G > A (D) G > H > A

Q20. If ${}^nP_2 = 30$, then n = :

(A) 6 (B) 4

(C) 5 (D) 8

Mathematics (Subjective) Group - I

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 1109-XI123-16000

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) Express the complex number $1 + i\sqrt{3}$ in polar form.

(ii) Whether closed or NOT with respect to addition and multiplication is {1}?

(iii) If C = {a, b, c, d}, find P(C).

(iv) Let U = the set of English alphabet A = {x | x is a vowel}, B = {y | y is consonant}, verify DeMorgan's laws for these sets.

(v) Construct the truth table for statement $\sim(p \rightarrow q) \leftrightarrow (p \wedge \sim q)$

(vi) Find the matrix A if, [[5, -1], [0, 0], [3, 1]] A = [[3, -7], [0, 0], [7, 2]]

(vii) If A = [a_ij]_(3×4), then show that I_3 A = A

(viii) Without expansion verify that $|mn, l, l^2; nl, m, m^2; lm, n, n^2| = |1, l^2, l^3; 1, m^2, m^3; 1, n^2, n^3|$

(ix) Find roots of equation by quadratic formula $(x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0$

(x) Find four fourth roots of 625.

(xi) Find the condition that $a/(x-a) + b/(x-b) = 5$ may have roots equal in magnitude but opposite in sign.

(xii) Show that roots of equation $px^2 - (p-q)x - q = 0$ will be rational.

Q3. (Marks 16) Attempt any EIGHT parts:

(i) Define partial fraction resolution.

(ii) If 1/a, 1/b and 1/c are in A.P, show that the common difference is (a-c)/(2ac)

(iii) Find A.M between $1 - x + x^2$ and $1 + x + x^2$.

(iv) How many terms of series -7 + (-5) + (-3) + ... amount to 65?

(v) Find vulgar fraction equivalent to the recurring decimal 1.34

(vi) If 5 is the H.M between 2 and b, find b.

(vii) How many arrangements of the letters of the word "MATHEMATICS", taking all together, can be made?

(viii) Find the value of n when ${}^nC_{10} = (12\times11)/(2!)$

(ix) There are 5 green and 3 red balls in a box. One ball is taken out, find the probability that the ball taken out is red.

(x) Prove that $n! > 2^{n-1}$ for n = 4, 5.

(xi) Find 6th term in the expansion of $(x^2 - 3/(2x))^{10}$.

(xii) Write first 4 terms of the expansion of $(8 - 2x)^{-1}$

Q4. (Marks 18) Attempt any NINE parts:

(i) Find l, when $\theta = 65°20'$, r = 18mm

(ii) Verify $\cos2\theta = \cos^2\theta - \sin^2\theta$ when $\theta = 30°, 45°$

(iii) Prove the identity $(2\tan\theta)/(1 + \tan^2\theta) = 2\sin\theta\cos\theta$

(iv) Prove that $\sin(180° + \alpha)\sin(90° - \alpha) = -\sin\alpha\cos\alpha$

(v) Prove that $\tan(\pi/4 - \theta) + \tan(3\pi/4 + \theta) = 0$

(vi) Prove the identity $(\sin3\theta)/(\sin\theta) - (\cos3\theta)/(\cos\theta) = 2$

(vii) Define period of a trigonometric function.

(viii) A vertical pole is 8m high and the length of its shadow is 6m. What is the angle of elevation of the sun at that moment?

(ix) Find the area of the triangle ABC, in which b = 21.6, c = 30.2 and $\alpha = 52°40'$

(x) Prove that with usual notations $r\, r_1\, r_2\, r_3 = \Delta^2$

(xi) Show that $\cos(2\sin^{-1}x) = 1 - 2x^2$

(xii) Solve the equation $\sin x + \cos x = 0$

(xiii) Solve the trigonometric equation $\operatorname{cosec}^2\theta = 4/3$ in $[0, 2\pi]$

> Note: SECTION - II

> Note: Attempt any THREE questions. Each question carries 10 marks.

Q5.

(a) [05] Solve the following system of linear equations by Cramer's rule: $2x_1 - x_2 + x_3 = 5$; $4x_1 + 2x_2 + 3x_3 = 8$; $3x_1 - 4x_2 - x_3 = 3$

(b) [05] If α and β are the roots of $x^2 - 3x + 5 = 0$, form the equation whose roots are $(1-\alpha)/(1+\alpha)$ and $(1-\beta)/(1+\beta)$

Q6.

(a) [05] Resolve into partial fractions: $x^4/(1 - x^4)$

(b) [05] Prove that ${}^{n-1}C_r + {}^{n-1}C_{(r-1)} = {}^{n}C_r$

Q7.

(a) [05] Obtain the sum of all integers in the first 1000 integers which are neither divisible by 5 nor by 2

(b) [05] If $y = 2/5 + (1.3)/(2!)(2/5)^2 + (1.3.5)/(3!)(2/5)^3 + ...$, then show that $y^2 + 2y - 4 = 0$

Q8.

(a) [05] Find the values of all trigonometric functions of $19\pi/3$

(b) [05] Reduce $\sin^4\theta$ to an expression involving only functions of multiples of θ raised to the first power.

Q9.

(a) [05] Prove that $r = \Delta/s$ with usual notations.

(b) [05] Prove that $2\tan^{-1}(2/3) = \sin^{-1}(12/13)$

GROUP – II

Mathematics (Objective) Group - II

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6198

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If n = 1, then value of n(n-1)! is:

(A) 0 (B) 2

(C) 1 (D) -1

Q2. $(n^3 + 2n)/3$ represents an / a:

(A) Real number (B) Rational number

(C) Irrational number (D) Integer

Q3. In the expansion of $(a + b)^7$, the 2nd term is:

(A) a^7 (B) $7ab^6$

(C) $7a^6b^6$ (D) $7a^6b$

Q4. 1° = :

(A) 180/π radian (B) π/180 radian

(C) 1/(180π) radian (D) 180π radian

Q5. sin3x = :

(A) $4\cos^3x - 3\cos x$ (B) $3\cos^3x - 4\cos x$

(C) $3\sin x - 4\sin^3x$ (D) $4\sin x - 3\sin^3x$

Q6. Range of sin x is:

(A) [-1, 1] (B) (-1, 1)

(C) [-1, 1) (D) (-1, 1]

Q7. If ABC be any triangle and γ = 90°, then:

(A) $a^2 - c^2 = b^2$ (B) $a^2 + b^2 + c^2 = 0$

(C) $a^2 + c^2 = b^2$ (D) $a^2 + b^2 = c^2$

Q8. With usual notation area of triangle ABC is:

(A) √(s-a)(s-b)(s-c) (B) b/(a sin β)

(C) ab sin γ (D) (ac sin β)/2

Q9. $\tan(\tan^{-1}(-1)) =$:

(A) π/4 (B) -π/4

(C) -1 (D) 1

Q10. If sin x = 1/2, then x = :

(A) -π/6, 5π/6 (B) -π/6, 5π/6

(C) π/6, 5π/6 (D) π/3, 2π/3

Q11. The set, S = {1} is closed under:

(A) Multiplication (B) Addition

(C) Subtraction (D) Division

Q12. p ∧ ~q is a / an:

(A) Absurdity (B) Tautology

(C) Inverse (D) Converse

Q13. If A = [[x, 1], [1, 1]] and |A| = 1, then x = :

(A) 3 (B) 1

(C) 2 (D) 0

Q14. The matrix [a b c d] is a _____ matrix.

(A) Square (B) Row

(C) Unit (D) Null

Q15. For what value of K will equation $x^2 - Kx + 4 = 0$ have sum of roots equal to product of roots?

(A) 3 (B) -2

(C) -4 (D) 4

Q16. Which is a factor of $x^n + a^n$, n ∈ O?

(A) x + a (B) x - a

(C) x = 0 (D) x - 2a

Q17. $p(x)/(x^2 + 1)$ will be proper fraction if degree of p(x) = :

(A) 1 (B) 4

(C) 2 (D) 3

Q18. No, term of the geometric sequence is:

(A) 1 (B) 0

(C) 2 (D) -1

Q19. If two numbers are $3\sqrt{5}$ and $5\sqrt{5}$, then arithmetic mean will be:

(A) $8\sqrt{5}$ (B) $4\sqrt{5}$

(C) $2\sqrt{5}$ (D) $6\sqrt{5}$

Q20. ${}^{6}P_{4}$ = :

(A) 3450 (B) 361

(C) 360 (D) 363

Mathematics (Subjective) Group - II

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 1110-XI123-23000

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) State trichotomy property of real numbers.

(ii) Express i/(1 + i) in the form of a + bi

(iii) Write the descriptive and tabular form of set A = {x : x ∈ E ∧ 4 ≤ x ≤ 10}

(iv) Find the converse and inverse of q → p

(v) Define group.

(vi) Find x and y if [[x+3, 1], [-3, 3y-4]] = [[2, 1], [-3, 2]]

(vii) If A and B are non-singular matrices, then show that $(AB)^{-1} = B^{-1}A^{-1}$

(viii) Let A = [[1, 2, 0], [3, 2, -1], [-1, 3, 2]], show that A + A^t is symmetric.

(ix) Find three cube roots of unity.

(x) Show that x + a is a factor of $x^n + a^n$ where n is odd integer.

(xi) If α and β are roots of $5x^2 - x - 2 = 0$, then find value of 3/α + 3/β.

(xii) Show that roots of equation $(p + q)x^2 - px - q = 0$ are rational.

Q3. (Marks 16) Attempt any EIGHT parts:

(i) $(3x^2 + 1)/(x - 2)$ is an improper fraction, convert into proper fraction.

(ii) Find a_8 for the sequence 1, 1, -3, 5, -7, 9, ...

(iii) Sum the series (-3) + (-1) + 1 + 3 + ... + a_{16}

(iv) Find the nth term of H.P. 1/2, 1/5, 1/8 .....

(v) Find the sum of infinite geometric series 1/2 + 1/4 + 1/8 + ...

(vi) Define arithmetic progression.

(vii) Evaluate 8!/6!

(viii) How many signals can be made with 4-different flags when any number of them are to be used at a time?

(ix) Find the value of n when ${}^{n}C_5 = {}^{n}C_4$

(x) Expand $(a/2 - 2/a)^6$

(xi) Find the term independent of x in the expansion of $(x/2 + 2/x^2)^{12}$

(xii) Using binomial theorem, find the value of $\sqrt{99}$ up to three places of decimals.

Q4. (Marks 18) Attempt any NINE parts:

(i) Find θ, when l = 3.2m, r = 2m

(ii) Prove the identity $\cot^4\theta + \cot^2\theta = \operatorname{cosec}^4\theta - \operatorname{cosec}^2\theta$

(iii) For $\theta = (-71/6)\pi$, find the values of sin θ and cos θ

(iv) Show that $\cos330°\sin600° + \cos120°\sin150° = -1$

(v) Prove that $\sin2\alpha = 2\sin\alpha\cos\alpha$

(vi) Express $2\sin7\theta\sin2\theta$ as a sum or difference.

(vii) Find the period of sin(x/5)

(viii) In the triangle ABC if a = 36.21, c = 30.14, β = 78°10' find angle γ

(ix) The area of triangle is 2437. If a = 79, c = 97 find angle β.

(x) Show that $r_2 = s\tan(\beta/2)$

(xi) Without using table / calculator show that $\cos^{-1}(4/5) = \cot^{-1}(4/3)$

(xii) Find the solutions of cosec x = 2, $x \in [0, 2\pi]$

(xiii) Solve sin x + cos x = 0

> Note: SECTION - II

> Note: Attempt any THREE questions. Each question carries 10 marks.

Q5.

(a) [05] Solve the system of linear equations by Cramer's rule: $2x_1 - x_2 + x_3 = 8$; $x_1 + 2x_2 + 2x_3 = 6$; $x_1 - 2x_2 - x_3 = 1$

(b) [05] Solve the equation: $x^2 - x/2 - 7 = x - 3\sqrt{(2x^2 - 3x + 2)}$

Q6.

(a) [05] Resolve into partial fractions: $2x^4/((x - 3)(x + 2)^2)$

(b) [05] Find value of n and r when ${}^{n-1}C_{(r-1)} : {}^{n}C_r : {}^{n+1}C_{(r+1)} = 3:6:11$

Q7.

(a) [05] The sum of three numbers in A.P is 24 and their product is 440. Find the numbers.

(b) [05] If x is so small that its square and higher powers can be neglected, then show that: (1+x)^(1/2)(4+3x)^(3/2)/(8+5x)^(1/3) ≈ 4(1 - 5x/6)

Q8.

(a) [05] Prove the identity $(\cos\theta + \sin\theta)/(\cos\theta - \sin\theta) + (\cos\theta - \sin\theta)/(\cos\theta + \sin\theta) = 2/(1 - 2\sin^2\theta)$

(b) [05] Prove that $(\sin\theta + \sin3\theta + \sin5\theta + \sin7\theta)/(\cos\theta + \cos3\theta + \cos5\theta + \cos7\theta) = \tan4\theta$

Q9.

(a) [05] Prove that $r_3 = s \tan(\gamma/2)$

(b) [05] Prove that $\sin^{-1}(4/5) + \sin^{-1}(5/13) + \sin^{-1}(16/65) = \pi/2$